Varendra University

Dept. of Computer Science & Engineering (CSE)

Arrays and Addressing Modes (Chapter-10)

Course Title: **Microprocessor and Assembly Language**
Course No: **CSE 3205**

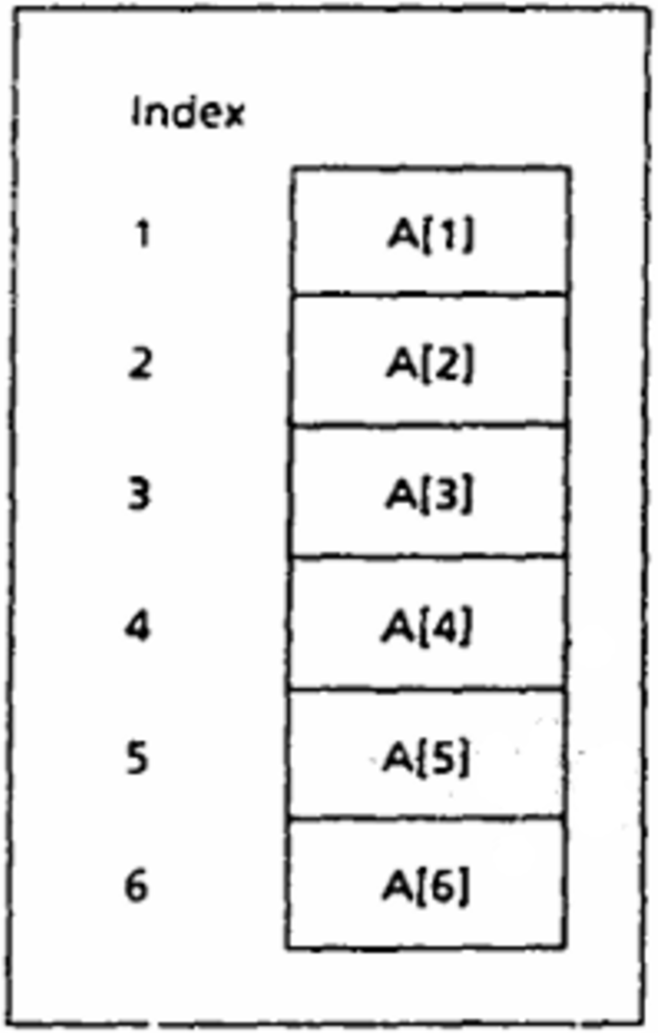
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Outline

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2. Two Dimensional Array.
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One Dimensional Array



One Dimensional Array

In Chapter 4, we used the DB and DW pseudo-ops to declare byte and word arrays; for example, a five-character string named MSG,

```
MSG   DB      'abcde'
```

or a word array W of six integers, initialized to 10,20,30,40,50,60.

```
W.    DW      10,20,30,40,50,60
```

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One Dimensional Array

Example 10.1: Exchange the 10th and 25th elements in a word array W.

Solution:

W[10] is located at address W + 9 × 2 = W + 18 and W[25] is at W + 24 × 2 = W + 48; so we can do the exchange as follows:

```
MOV   AX,W+18          ;AX has W[10]
XCHG  W+48,AX          ;AX has W[25]
MOV   W+18,AX          ;complete exchange
```

One Dimensional Array

Example 10.3: Write some code to sum in AX the elements of the 10-element array W defined by,
W DW 10, 20, 30, 40, 50, 60, 70, 80, 90, 100

Solution:

```
        XOR   AX,AX             ;AX holds sum
        LEA   SI,W              ;SI points to array W
        MOV   CX,10             ;CX has number of elements
ADDNOS:
        ADD   AX,[SI]           ;sum = sum + element
        ADD   SI,2              ;move pointer to the next
                                ;element
        LOOP ADDNOS             ;loop until done
```

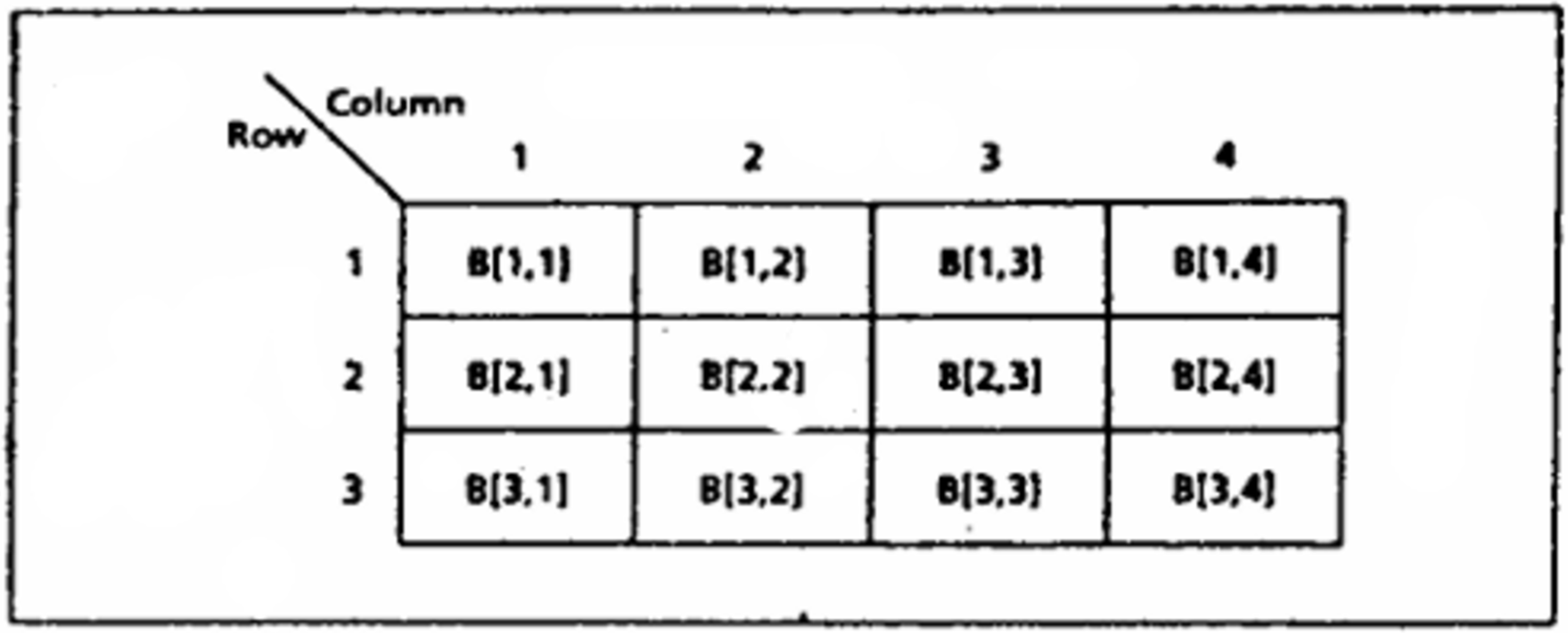
One Dimensional Array

Example 10.7(*): Replace each lowercase letter in the following string by its upper case equivalent. Use index addressing mode:
MSG DB 'this is a message'

Solution:

```
        MOV   CX,17             ;no. of chars in string
        XOR   SI,SI             ;SI indexes a char
TOP:
        CMP   MSG[SI],' '       ;blank?
        JE    NEXT              ;yes, skip over
        AND   MSG[SI],0DFh      ;no, convert to upper case
NEXT:
        INC   SI                ;index next byte
        LOOP  TOP               ;loop until done
```

Two Dimensional Array



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One Dimensional Array

Example 10.12: Suppose A is an M × N word array stored in row-major order.

1. Where does row i begin?
2. Where does column j begin?
3. How many bytes are there between elements in a column?

Solution:

1. Row i begins at A[i, 1]; by formula (1) its address is A + (i – 1) × *N* × 2.
2. Column j begins at A[1, j]; by formula (1) the address is A + (j – 1) × 2.
3. Because there are *N* columns, there are 2 × *N* bytes between elements in any given column.

Assignment

Write a code in assembly language which will reverse an array of N words.

Source:
Example **10.4** (Chapter 10).

Book: Assembly Language Programming and Organization of the IBM PC.
Writer: Ytha Yu, Charles Marut.

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Assignment

Suppose a class of five students is given four exams. The results are recorded as follows:

	Test 1	Test 2	Test 3	Test 4
MARY ALLEN	67	45	98	33
SCOTT BAYLIS	70	56	87	44
GEORGE FRANK	82	72	89	40
BETH HARRIS	80	67	95	50
SAM WONG	78	76	92	60

We will write a program to find the class average on each exam. To do this, we sum the entries in each column and divide by 5.

Source:

Example **10.6** (Chapter 10).

Book: Assembly Language Programming and Organization of the IBM PC.

Writer: Ytha Yu, Charles Marut.